

Jay Knight

Reno, NV (Open to Relocation Nationwide) | jrk.reno@gmail.com | (775)762-0792 | github.com/jrkre

EDUCATION University of Nevada, Reno

B.S. Computer Science and Engineering | *Minor in Mathematics*

Expected 2026

COURSEWORK Algorithms & Data Structures, Systems Administration, Database Management, Embedded Systems, Motion Planning, Mobile Robots

EXPERIENCE Undergraduate Researcher | University of Nevada, Reno

Aug 2023 - Present

- Designed and implemented modular, configurable software systems using ROS 2, enabling non-technical users to tailor system behavior without code changes.
- Built and deployed a full-stack web application (Flask, simulation backend, real-time data pipelines) supporting research workflows and stakeholder feedback.
- Developed and maintained C++ and Python services on embedded Linux, supporting multiple hardware platforms and long-term system reliability.
- Collaborated with interdisciplinary stakeholders to gather requirements, iterate on system design, and support deployed solutions.

REU Undergraduate Researcher | University of Nevada, Reno

Jun 2023 - Aug 2023

- Analyzed complex workflows and translated user needs into technical system architecture for human-robot collaborative tasks.
- Worked independently in a fast-paced research environment with shifting priorities and deadlines.

K-12 Robotics Education Assistant | Nevada Robotics | Reno, NV

Nov 2021 - Aug 2024

- Served as a technical liaison between educators, administrators, and engineering teams across Nevada.
- Trained 500+ teachers statewide, delivering clear technical explanations to non-technical audiences.
- Led modernization of a statewide robotics lending and tracking system, improving reporting accuracy and operational efficiency.
- Maintained data integrity and confidentiality for statewide educational programs.

PUBLICATIONS Knight, J. et al., *Design Activity for Personalized Robot Faces: Evaluating Child Responses to Customizable Expressive Faces*, IEEE/ACM RO-MAN 2025.

PROJECTS **Bloom** | Serving as technical lead for a senior capstone project, owning the full lifecycle from requirements elicitation to agile execution and delivery.

HomeLab | Designed and maintained secure Linux-based systems emphasizing reliability, uptime, and low-latency access. Automated deployment and configuration for reproducible environments, mirroring firmware build and test pipelines.

'RPI Robot' | Designed and implemented a cost-effective embedded robotic system using C/C++ on Linux-based hardware. Implemented real-time task scheduling and concurrency control.

LEADERSHIP UNR Robotics Club | President, Co-Founder

Dec 2022 - May 2024

- Co-founded and secured \$10k in grants for education and team development; created community resources for K-12 robotics teams.

FBLA Nevada State Champion | Mobile Application Development

May 2018

- Designed and delivered a mobile application solution evaluated against business and technical criteria.

SKILLS **Systems & Platforms:** Linux, MacOS, Windows, ROS 2, Embedded Linux, Azure Deployments, Systems Administration

Programming: C++, Python, .NET Core, NodeJS, Bash, Git

Robotics & Hardware: LiDAR, Sensors/Actuators, Robot Fabrication, PCB/Mechanical Assembly, Motion Planning, Additive Manufacturing

Other: CAD, Blender, Technical Writing, Project Management, Interdisciplinary Collaboration